

Table 1. Summary Statistics for Plot-Level Analysis

	Obs	Mean	Std. Dev.	Min	Max
Wildfire occurrence	761250	.055	.227	0	1
Harvest	761250	.011	.103	0	1
Thinning	761250	.006	.08	0	1
Harvest shift-share	761250	.013	.009	0	.172
Thinning shift-share	761250	.007	.011	0	.076
Total forest area (thousand acre)	761250	5.83	3.213	.001	44.892
Maximum VPD (hPa)	761250	19.079	4.071	9.95	44.3
Trees per acre	761250	328.469	259.092	0	999.662
Sawtimber volume per acre (ft ³ /acre)	761250	699.04	1031.797	0	17464.57
Stand age	761250	40.991	26.112	0	200
Percentage artificial forest	761250	.146	.33	0	1
Slope (%)	761250	8.832	14.438	0	155
Elevation (m)	761250	752.652	800.871	0	7170
Bio growth rate	761250	.967	1.648	-146.259	55.621
Forestry and logging establishments	761250	5.233	7.459	0	55
Distance to nearest road (mile)	761250	.394	.553	0	6
Siteclass (Best:1 to Worst:7)	761250	5.103	1.294	1	7

Notes: Summary statistics are reported for a plot-year panel covering 37071 plots observed from 2000 to 2024, summing up to 761,250 observations. Stand age is capped at 200 years in the FIA data; 5% of plots have an average stand age greater than 82 years, and 1% have an average stand age greater than 101 years. Bio growth rate is the annual aboveground biomass growth rate of a stem (short tons/ year). Site productivity captures the inherent ability of forest land to support timber growth, defined by potential wood production (ft³/acre/year) under fully stocked conditions.

Table 2. Summary Statistics for County-Level Analysis

	Mean	Std. Dev.	Min	Max
Wildfire area in MTBS (thousand acre)	.753	6.972	0	348.916
Wildfire occurrence in MTBS	.092	.289	0	1
Harvest area (thousand acre)	2.449	5.179	0	70.224
Thinning area (thousand acre)	1.472	4.024	0	47.844
Harvest shift-share (thousand acre)	2.564	2.759	0	42.985
Thinning shift-share (thousand acre)	1.524	2.36	0	23.3
Maximum VPD (hPa)	31.774	9.246	14.311	78.167
Trees per acre	510.997	225.387	0	998.441
Total forest area (thousand acre)	209.12	148.844	.549	1637.981
Sawtimber volume per acre (ft ³ /acre)	870.075	682.519	0	11163.57
Bio growth rate	1.31	.951	-5.66	7.321
Slope (%)	10.143	11.665	0	65
Elevation (m)	798.063	807.996	0	5487.093
Stand age	42.738	13.793	0	135
Site productivity	66.975	31.648	9.5	225
Forestry and logging establishments	3.783	6.055	0	55

Notes: Summary statistics are reported for a balanced county-year panel covering 1,263 counties observed annually from 2000 to 2023. Bio growth rate is the annual aboveground biomass growth rate of a stem (short tons/ year). Site productivity captures the inherent ability of forest land to support timber growth, defined by potential wood production (ft³/acre/year) under fully stocked conditions. 73 counties excluded because of unrealistic trees per acre information.

Table 3. Direct mitigation benefits from fuel reduction treatment on wildfire risk

	Harvest		Thinning		Joint IV
	(1)	(2)	(3)	(4)	(5)
	Wildfire event FE	Wildfire event IV	Wildfire event FE	Wildfire event IV	Wildfire event IV
Harvest	-0.0000 (0.0014)	-0.4453*** (0.0830)			-0.3883*** (0.0824)
Thinning			-0.0014 (0.0018)	-0.1872*** (0.0435)	-0.1663*** (0.0471)
Kleibergen-Paap Wald F statistic		214.64		488.11	107.14
Observations	761,236	761,236	761,236	761,236	761,236
Plot FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Species group FE	✓	✓	✓	✓	✓
Ecoregion FE	✓	✓	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The difference between harvest and thinning is statistically significant with p-value=0.034. 14 singleton observations are dropped in the IV models because they do not provide within-group variation after absorbing the fixed effects.

Table 4. Combined mitigation benefits from fuel reduction treatment on wildfire risk at county level

	Harvest		Thinning		Joint IV
	(1) FE	(2) IV	(3) FE	(4) IV	(5) IV
Harvest	-0.0036 (0.0048)	-0.4622*** (0.1138)			-0.2527** (0.1248)
Thinning			-0.0093 (0.0064)	-0.2107*** (0.0485)	-0.1673*** (0.0573)
Kleibergen-Paap Wald F statistic	.	59.61	.	137.53	21.17
Observations	30312	30312	30312	30312	30312
County FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The difference between harvest and thinning is not statistically significant with p-value= 0.600.

Table A1. Direct effects of fuel reduction treatment on wildfire risk

	Harvest			Thinning			Joint IV	Thinning vs. Wait
	(1) Wildfire event FE	(2) Harvest event First stage	(3) Wildfire event IV	(4) Wildfire event FE	(5) Thinning event First stage	(6) Wildfire event IV	(7) Wildfire event IV	(8) Wildfire event IV
Harvest	-0.0000 (0.0014)		-0.4453*** (0.0830)				-0.3883*** (0.0824)	
Thinning				-0.0014 (0.0018)		-0.1872*** (0.0435)	-0.1663*** (0.0471)	0.2166** (0.0978)
No treatment / wait								0.3871*** (0.0822)
Harvest shift-share instrument		0.6116*** (0.0417)						
Thinning shift-share instrument					0.7804*** (0.0353)			
Maximum VPD	0.0118*** (0.0007)	0.0001 (0.0001)	0.0118*** (0.0007)	0.0118*** (0.0007)	0.0001 (0.0001)	0.0117*** (0.0007)	0.0118*** (0.0007)	0.0118*** (0.0007)
Trees per acre	0.0005** (0.0002)	-0.0210*** (0.0006)	-0.0089*** (0.0018)	0.0005** (0.0002)	-0.0037*** (0.0004)	-0.0002 (0.0003)	-0.0083*** (0.0017)	-0.0083*** (0.0017)
Sawtimber volume per acre	-0.0006 (0.0009)	-0.0092*** (0.0016)	-0.0044*** (0.0014)	-0.0006 (0.0009)	-0.0041*** (0.0014)	-0.0013 (0.0010)	-0.0046*** (0.0013)	-0.0047*** (0.0013)
Stand age	-0.0006 (0.0005)	-0.0099*** (0.0008)	-0.0051*** (0.0010)	-0.0006 (0.0005)	0.0087*** (0.0005)	0.0010 (0.0006)	-0.0031*** (0.0011)	-0.0031*** (0.0011)
Slope	-0.0001 (0.0001)	0.0005*** (0.0001)	0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
Elevation	0.0044 (0.0059)	-0.0148** (0.0060)	-0.0021 (0.0065)	0.0044 (0.0059)	-0.0007 (0.0006)	0.0043 (0.0060)	-0.0014 (0.0064)	-0.0014 (0.0064)
Bio growth rate	-0.0010*** (0.0004)	-0.0012*** (0.0004)	-0.0017*** (0.0004)	-0.0010*** (0.0004)	0.0057*** (0.0009)	0.0002 (0.0005)	-0.0005 (0.0005)	-0.0005 (0.0005)
Forestry establishments	0.0008** (0.0004)	-0.0001 (0.0002)	0.0007* (0.0004)	0.0008** (0.0004)	-0.0002 (0.0002)	0.0006* (0.0004)	0.0006 (0.0004)	0.0006 (0.0004)
Kleibergen-Paap Wald F statistic			214.64			488.11	107.14	108.03
Observations	761,236	761,236	761,236	761,236	761,236	761,236	761,236	761,236
Plot FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Species group FE	✓	✓	✓	✓	✓	✓	✓	✓
Ecoregion FE	✓	✓	✓	✓	✓	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: 14 singleton observations are dropped in all models because they do not provide within-group variation after absorbing the fixed effects.

Table A2. Robustness tests with alternative treatment windows

	Harvest			Thinning			Joint IV		
	(1) Wildfire event	(2) Wildfire event	(3) Wildfire event	(4) Wildfire event	(5) Wildfire event	(6) Wildfire event	(7) Wildfire event	(8) Wildfire event	(9) Wildfire event
Harvest (3-year)	-0.4859*** (0.1022)						-0.4018*** (0.1072)		
Harvest (4-year)		-0.4453*** (0.0830)						-0.3883*** (0.0824)	
Harvest (5-year)			-0.4327*** (0.0751)						-0.3980*** (0.0713)
Thinning (3-year)				-0.3295*** (0.0603)			-0.2610*** (0.0667)		
Thinning (4-year)					-0.1872*** (0.0435)			-0.1663*** (0.0471)	
Thinning (5-year)						-0.0793** (0.0338)			-0.0891** (0.0377)
Kleibergen-Paap Wald F statistic	217.04	214.64	191.47	418.96	488.11	557.45	93.84	107.14	105.77
Observations	774,295	761,236	746,245	774,295	761,236	746,245	774,295	761,236	746,245
Plot FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Species group FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ecoregion FE	✓	✓	✓	✓	✓	✓	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: The treatment variables are defined using alternative lag windows over the previous three, four, and five years. The 4-year-window models, reported in columns (2), (5), and (8), include 761,236 observations; compared with the 3-year-window models, 13,059 observations are dropped because the required longer lagged treatment information is unavailable. The 5-year-window models, reported in columns (3), (6), and (9), include 746,245 observations; compared with the 3-year-window models, 28,050 observations are dropped for the same reason. Across specifications, the estimated coefficients are larger in absolute magnitude for the shorter treatment windows for both harvest and thinning. This pattern indicates that the wildfire-mitigation effects of forest treatments are strongest in the years immediately following treatment and gradually diminish as the time since treatment increases. Such attenuation is consistent with the gradual reaccumulation of vegetation and combustible fuels following harvest or thinning, which can reduce the effectiveness of earlier treatments in mitigating subsequent wildfire risk.

Table A3. Robustness tests with alternative IV

	Harvest			Thinning		
	(1) State IV	(2) State-Eco IV	(3) County IV	(4) State IV	(5) State-Eco IV	(6) County IV
Harvest	-0.4453*** (0.0830)	-0.2790*** (0.0685)	-0.1950*** (0.0750)			
Thinning				-0.1872*** (0.0435)	-0.1831*** (0.0437)	-0.1065** (0.0530)
Kleibergen-Paap Wald F statistic	214.64	262.55	155.54	488.11	415.47	165.12
Observations	761,236	761,236	761,236	761,236	761,236	761,236
Plot FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Species group FE	✓	✓	✓	✓	✓	✓
Ecoregion FE	✓	✓	✓	✓	✓	✓

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Columns (1)–(3) report IV estimates for harvest using delocalized shift-share instruments constructed at three levels of geographic aggregation: state, state-ecoregion, and county-ecoregion, respectively. Columns (4)–(6) report the corresponding estimates for thinning using the same set of instruments.

Table A4. Plot-level robustness tests without Texas and Oklahoma

	Harvest		Thinning	
	(1) Wildfire event	(2) Wildfire event	(3) Wildfire event	(4) Wildfire event
Harvest	-0.4453*** (0.0830)	-0.4585*** (0.0963)		
Thinning			-0.1872*** (0.0435)	0.0092 (0.0450)
Kleibergen-Paap Wald F statistic	214.64	181.85	488.11	438.74
Observations	761,236	542,064	761,236	542,064
Plot FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Species group FE	✓	✓	✓	✓
Ecoregion FE	✓	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Columns (1) and (3) use the full estimation sample. Columns (2) and (4) exclude Texas and Oklahoma, where artificially regenerated forests are limited, to reduce concerns that the shift-share instruments may rely on extrapolation beyond areas with adequate variation in artificial regeneration.

Table A5. Robustness tests for alternative wildfire indicators

	Log wildfire area (MTBS)	Wildfire event (MTBS)	Percent fire (MTBS)
Panel A. Harvest results	(1)	(2)	(3)
Log average harvest area	-0.462*** (0.114)	-0.061*** (0.015)	-0.001*** (0.000)
Kleibergen-Paap Wald F statistic	59.61	59.61	59.61
Observations	30312	30312	30312
County FE	✓	✓	✓
Year FE	✓	✓	✓
Panel B. Thinning results	Log wildfire area (MTBS)	Wildfire event (MTBS)	Percent fire (MTBS)
	(4)	(5)	(6)
Log average thinning area	-0.211*** (0.049)	-0.029*** (0.006)	-0.000 (0.000)
Kleibergen-Paap Wald F statistic	137.53	137.53	137.53
Observations	30312	30312	30312
County FE	✓	✓	✓
Year FE	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Percent wildfire is defined as the share of forest area burned by wildfire in that county-year. Because these measures normalize by county forest area, they reduce the influence of county size and show that the main results, especially the harvest treatment, are not solely driven by larger counties.

Table A6. Robustness of the Effects of Harvest and Thinning on County-Level Wildfire Burned Area

	Harvest						Thinning					
	Log wildfire area (1) FE	Log average treatment (2) First stage	Log wildfire area (3) IV: 4-year	Log wildfire area (4) IV: 3-year	Log wildfire area (5) IV: 5-year	Log wildfire area (6) IV no COVID	Log wildfire area (7) FE	Log average treatment (8) First stage	Log wildfire area (9) IV: 4-year	Log wildfire area (10) IV: 3-year	Log wildfire area (11) IV: 5-year	Log wildfire area (12) IV no COVID
Log average treatment	-0.0036 (0.0042)		-0.4622*** (0.1138)	-0.2916*** (0.0797)	-0.7207*** (0.1696)	-0.5702*** (0.1214)	-0.0093 (0.0063)		-0.2107*** (0.0485)	-0.2147*** (0.0427)	-0.1734** (0.0708)	-0.2692*** (0.0515)
Shift-share instrument		0.0001*** (0.0000)						0.0003*** (0.0000)				
Kleibergen-Paap Wald F statistic			59.61	57.78	42.83	60.47			137.53	141.86	88.92	132.73
Observations	30312	30312	30312	30312	29049	25260	30312	30312	30312	30312	29049	25260
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Columns (1)–(6) report estimates for harvest, and columns (7)–(12) report estimates for thinning. Columns (4) and (10) use a three-year treatment window, which reduces the sample by one year. Columns (5) and (11) use a five-year treatment window. Columns (6) and (12) exclude observations from year 2020, which reduces the sample by three years.